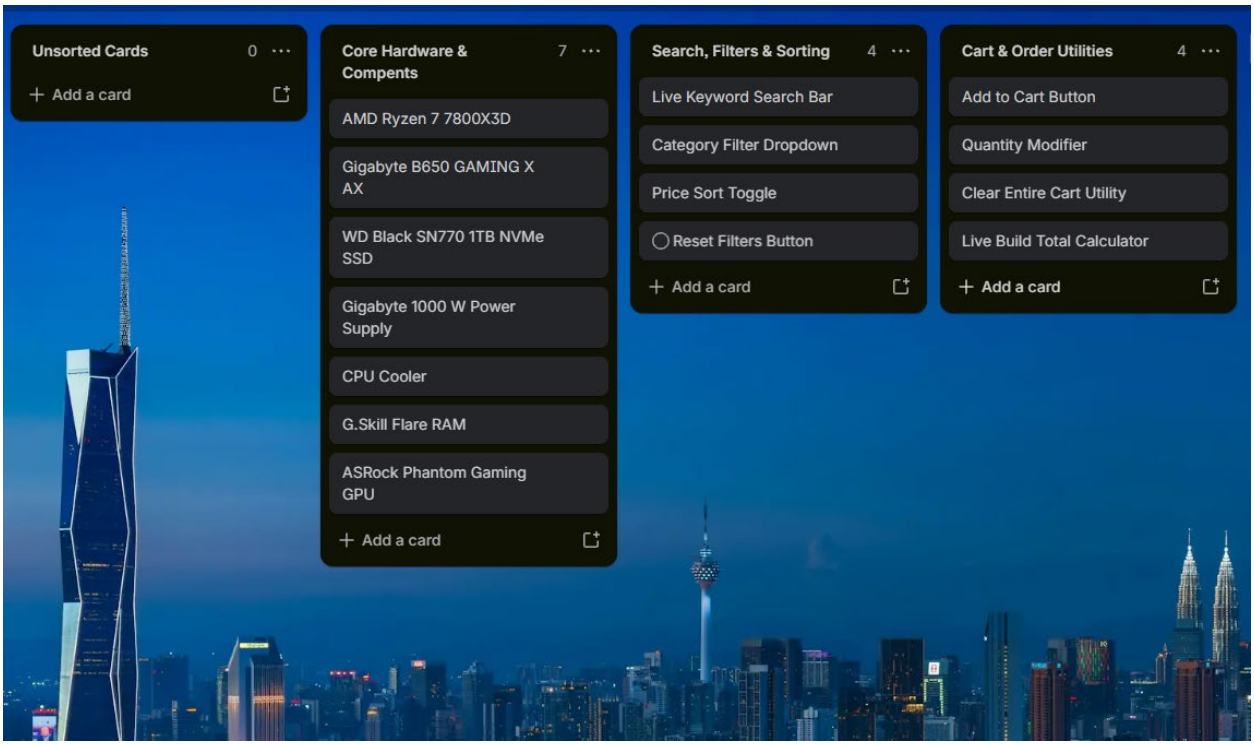
Empirical Card Sorting Experiment



Summary:

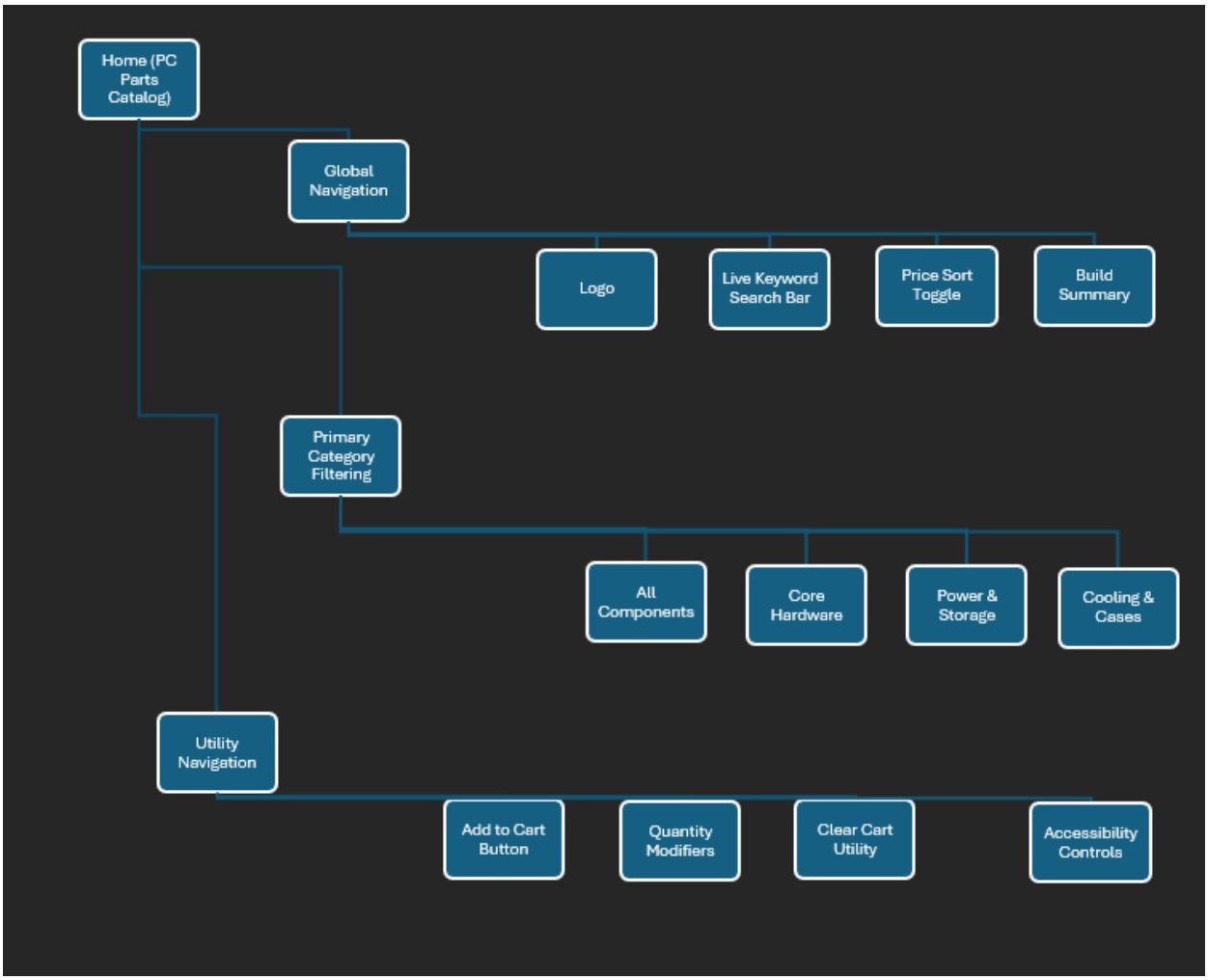
Participant Grouping Pattern: All 3 participants were able to sort these cards out into 3 main categories. A hardware category, searching category, and cart/total category.

Revised Sitemap & Information Architecture (IA)

Based on the participant groupings of the cards, the model moves from a rigid backend component list to a ordered out structure that divides the cards into 3 groups:

1. Global Navigation: Shows across all views to provide immediate access to the core system tools like live search and build summaries.
2. Primary Category Filtering: Single click access to filtered catalog items that are taken directly from the Hardware & Components sort.
3. Utility Navigation: Interactive modals handling cart state, quantities and accessibility overlays.

Sitemap



Low-Fidelity Wireframes & User Flows

1. Core Task Section
    - Flow A: Keyword Search & Category Filtering
    - Flow B: Cart Management & Quantity Modification
2. User Flow Diagramming

    Flow A:

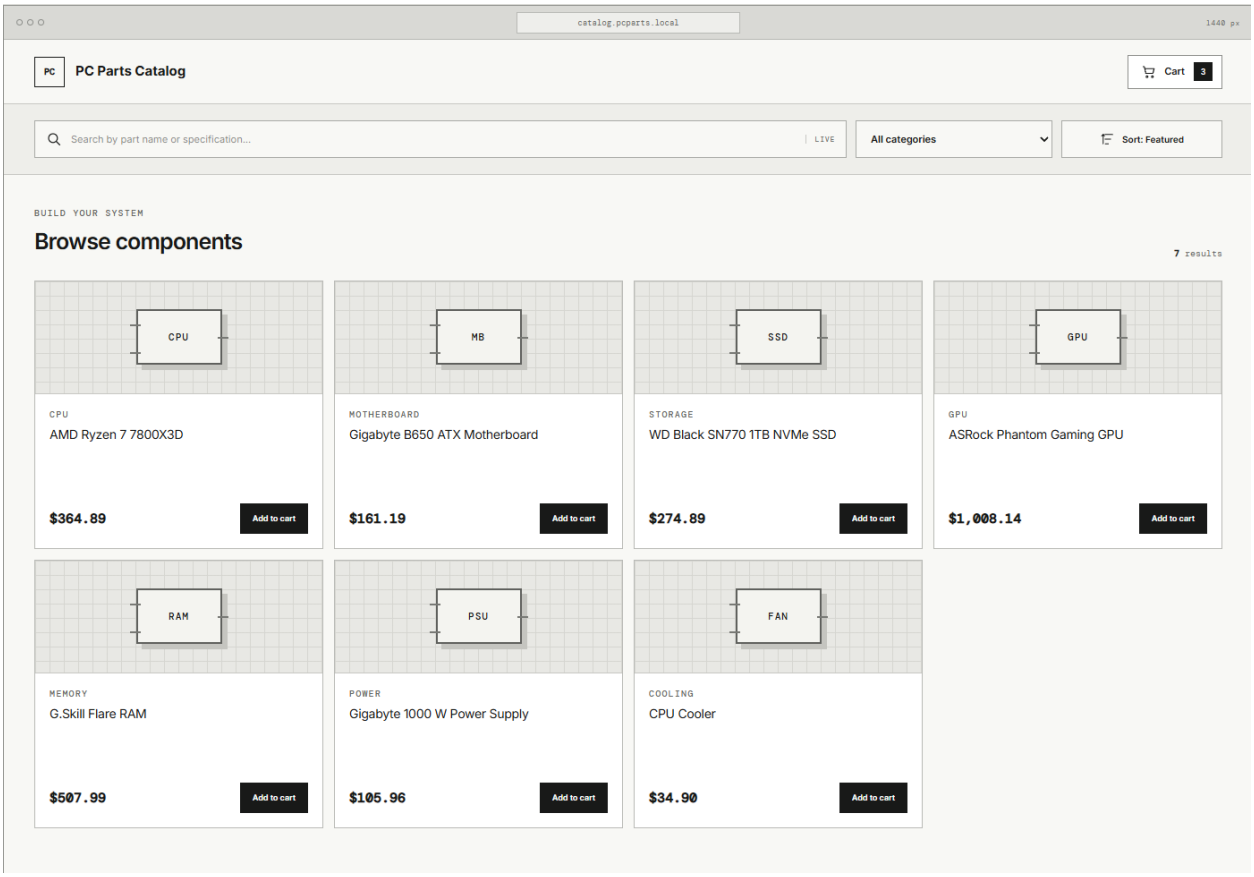
    - Step 1: The user starts on the default catalog view (home page) which shows all the core hardware components
    - Step 2: User chooses between utilizing the live keyword search bar or selecting an option from the category filter dropdown.
    - Step 3: The product grid updates dynamically to what the user selected. If a search yields no results then the system displays the empty state view.

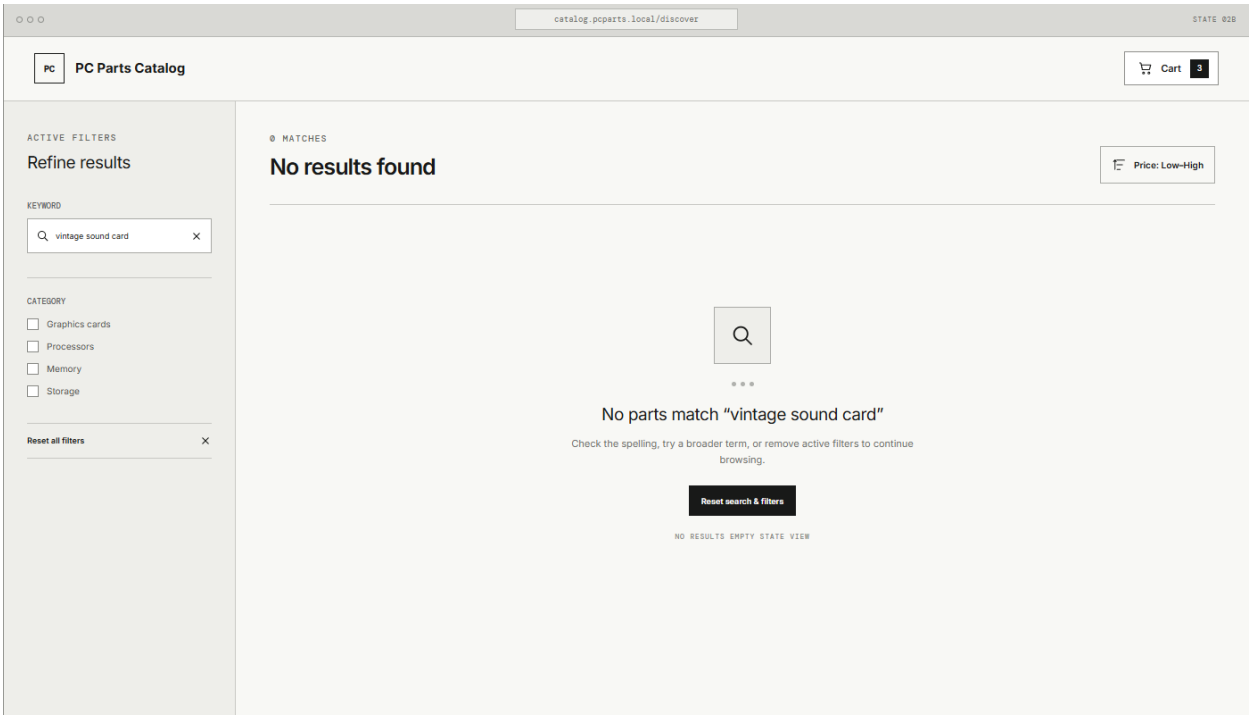
- Step 4: User interacts with the Price Sort Toggle to order the filtered components before selection.

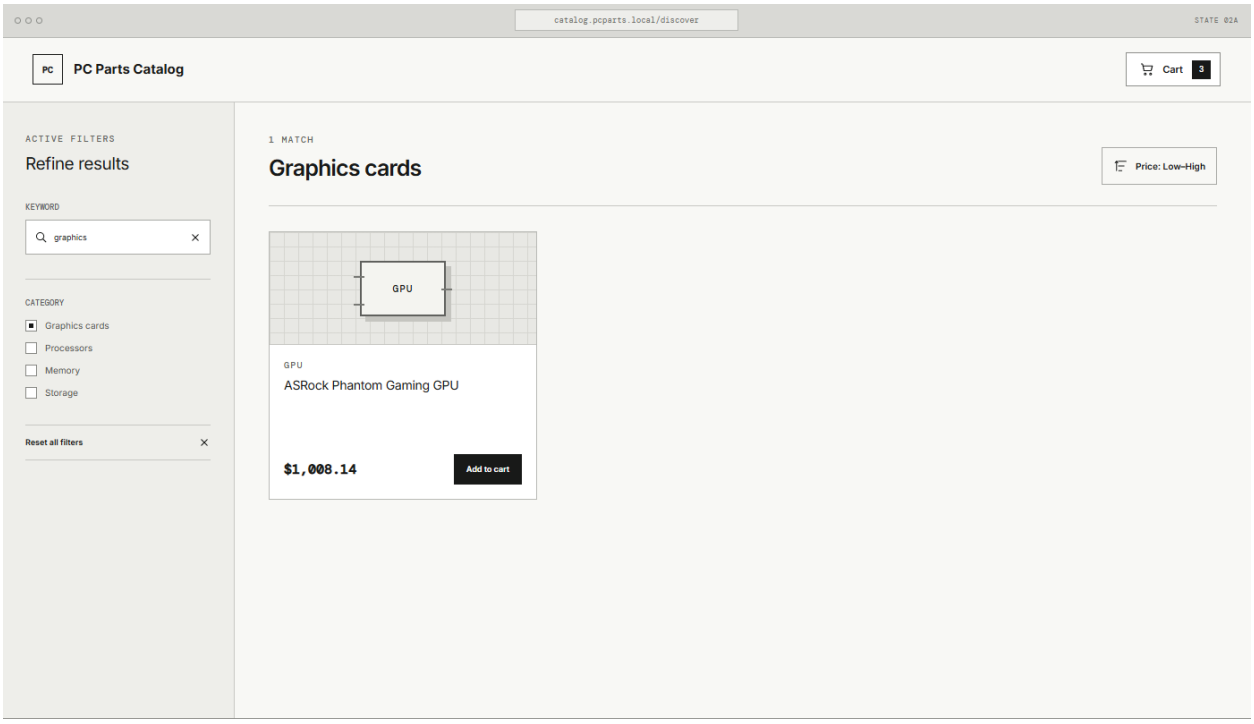
Flow B:

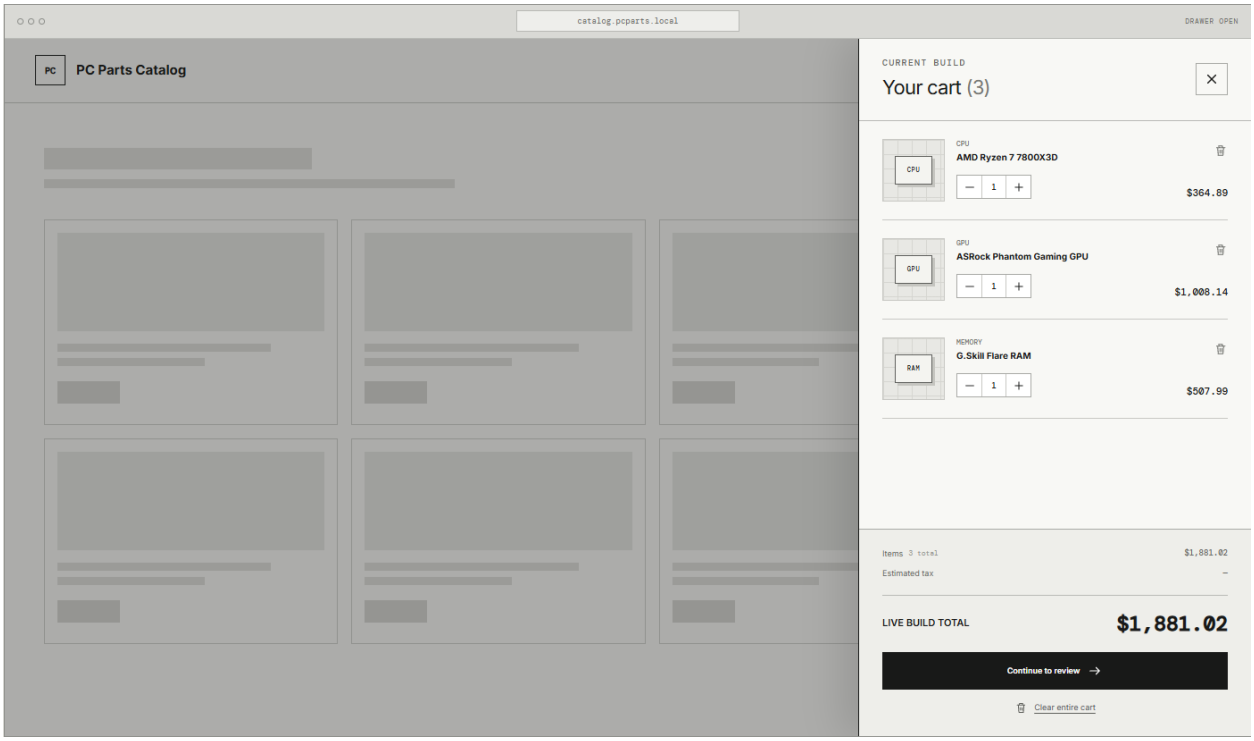
- Step 1: User clicks the Add to Cart Button on the hardware card they want.
- Step 2: The button triggers an immediate loading state, simulating asynchronous latency.
- Step 3: The item populates inside the side cart area and the Live Build Total Calculator updates dynamically.
- Step 4: The user clicks the quantity modifier to adjust the amount of the item(s) they want, or they click Clear Entire Cart Utility to clear the cart.

3. Wireframes Deck Creation




catalog.pcparts.local/discover
STATE 02B
PC
PC Parts Catalog
Cart 3
ACTIVE FILTERS
Refine results
KEYWORD
vintage sound card
CATEGORY
Graphics cards
Processors
Memory
Storage
Reset all filters
0 MATCHES
No results found
Price: Low-High
No parts match “vintage sound card”
Check the spelling, try a broader term, or remove active filters to continue browsing.
Reset search & filters
NO RESULTS EMPTY STATE VIEW


catalog.pcparts.local/discover
STATE 02A
PC
PC Parts Catalog
Cart 3
ACTIVE FILTERS
Refine results
KEYWORD
graphics
CATEGORY
Graphics cards
Processors
Memory
Storage
Reset all filters
1 MATCH
Graphics cards
Price: Low-High
GPU
GPU
ASRock Phantom Gaming GPU
$1,008.14
Add to cart



Accessibility First & Semantic Navigation Planning

1. Semantic HTML Structuring
    - <header>: Encloses the brand logo, title, and search controls
    - <nav>: Is home to primary category filters and sorting controls with those marked as “navigation”
    - <main>: Holds the core hardware product catalog grid and active filter state
    - <aside>: Contains the slide-out cart drawer and live build calculator
    - <footer>: Layout boundaries, copyright notes
    - <section>: Sections off individual wireframes and distinct UI modules
2. Keyboard & ARIA Blueprint
    - Standard Keyboard Traversal: Focus begins at the search input, moves through the category dropdown and sorting, moves to individual item cards, then finishes with the interactive add to cart button. All elements will have high-contrast visible indicator rings.
    - ARIA Attributes & State Management
        - role=”navigation” : Applied to the category filter block to have structured navigation points.
        - aria-current=”page” : Dynamically applied to the active category filter so that screen readers will announce the currently selected item.

- aria-expanded=”false/true” : This will be utilized in the cart trigger button and cart slide out container to tell the user if the cart overlay is collapsed or expanded.
- aria-live=”polite” : this will be configured on the live build total and item count regions to make sure that real time updates are announced without taking away the users focus.